UNIT - V
Robot Programming & Applications

Automation and Robotics (23A03606) - handwritten study notes

Syllabus Topics

- Robot Programming: methods of programming, requirements and features of programming languages, software packages, problems with programming languages
- Robot Application in Manufacturing: material transfer, material handling, loading and unloading
- Process applications: spot welding, continuous arc welding, spray painting
- Assembly and Inspection

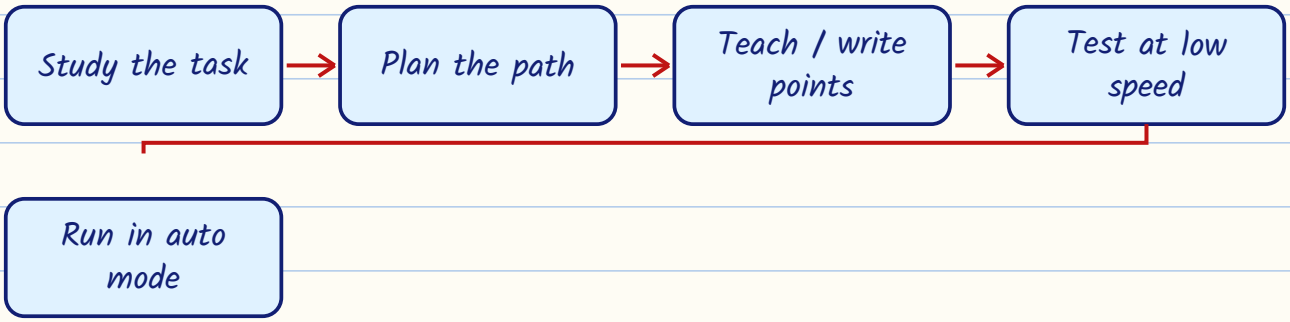
Exam tip: Unit V is mostly descriptive. Learn the three programming methods, the features of a good robot language, and one clear example of each application with its robot requirements.

1. Robot Programming - Introduction

A robot is useful only when it is **programmed**. Programming means teaching the manipulator the sequence of motions, positions, speeds and I/O actions needed to perform a task.

- A robot program is a list of instructions the controller executes in order
- It contains **motion commands** (go to a point), **position data** (coordinates, joint values), **sensor / I/O commands** (wait for signal) and **program-flow commands** (IF, GOTO, loops)
- The controller stores the program in memory and replays it, so the robot is called a **teach and playback** machine

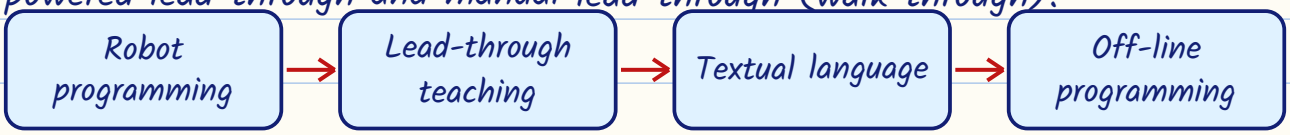
Programming workflow



A robot program = positions (where) + motions (how) + logic and I/O (when). All three must be defined.

2. Methods of Robot Programming

There are three broad approaches: **lead-through**, **textual (computer-like) languages** and **off-line programming**. Lead-through is further divided into powered lead-through and manual lead-through (walk-through).



2.1 Lead-through programming

The programmer **physically leads** the robot through the required motion and the controller records the points. During playback the robot repeats the recorded path.

(a) Powered lead-through (teach pendant)

- Uses a hand-held **teach pendant** with buttons / joystick / keys
- Each axis or the end effector is moved by keys in joint, world or tool coordinates
- When the arm reaches the desired position the programmer presses **RECORD / TEACH** and the point is stored
- Speed, function codes (open gripper, weld on) are added for each point
- Mostly used for **point-to-point** (PTP) tasks such as spot welding, pick and place

(b) Manual lead-through (walk-through)

- The programmer **grasps the arm** (or a dummy / light teach handle) and moves it manually through the complete path
- Controller samples the joint positions at a fixed time interval and stores them
- Suited for **continuous-path** tasks: spray painting, arc welding, polishing
- Robot must be light, or a **teach arm** (scaled replica) is used for heavy robots

Comparison of lead-through methods

Feature	Powered lead-through	Manual lead-through
Teaching device	Teach pendant	Operator moves the arm by hand
Path type	Point to point	Continuous path
Points stored	Selected key points	Many sampled points
Typical use	Spot welding, loading	Painting, seam welding
Accuracy of points	High	Depends on operator skill
Memory needed	Small	Large

Advantages of lead-through

- Easy to learn, no knowledge of a computer language is needed
- Programmer works with the real robot, so positions are exact for the real cell
- Quick for simple tasks

Disadvantages of lead-through

- Robot is **out of production** while being taught
- Difficult to add logic, branching, calculations and sensor decisions
- Hard to edit long programs and not easy to reuse for a similar part
- Danger to the programmer who works inside the work envelope

Teach mode safety: reduced speed (max 250 mm/s), enabling (dead-man) switch and emergency stop on the pendant.

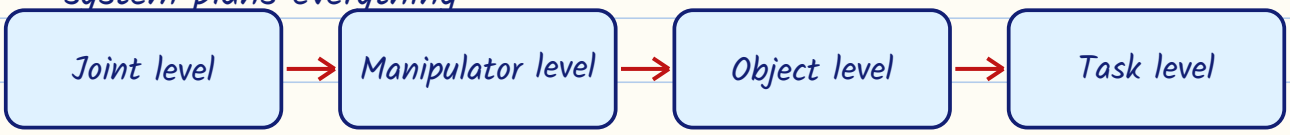
3. Textual (Computer-like) Robot Languages

In **textual programming** the programmer writes statements in a robot language using a keyboard and editor. The program is compiled or interpreted by the controller.

- Positions are defined as variables, e.g. frames and points, either taught once or calculated
- Supports **branching, loops, subroutines, arithmetic, sensor input and communication** with other machines
- Also called **explicit programming** or **robot-level programming**

Levels of robot programming languages

- **Joint level**: program the individual joint values
- **Manipulator level**: program the end effector position in Cartesian space (VAL, AL)
- **Object level**: command is on the objects (e.g. PLACE PART ON TABLE) and the system plans the motion
- **Task level**: only the final goal is given (e.g. ASSEMBLE GEARBOX); the system plans everything



> Higher level = easier for the user but needs a more powerful controller with world model and planner.

4. Some Popular Languages

Language	Developer	Remarks
VAL / VAL II	Unimation (PUMA)	Interpreter based, easy, widely taught
AL	Stanford University	Research, concurrent motions, force control
AML	IBM	Manufacturing language for IBM robots
RAPID	ABB	Modern structured language, very readable

KRL	KUKA	Used for KUKA robots
KAREL	FANUC	Pascal-like structured language
URScript	Universal Robots	Python-like for cobots

VAL style example (pick and place)

A short program in VAL-like syntax is shown below, written as plain lines:

- PROGRAM PICKPLACE
  - SPEED 50
  - MOVE P1 (go to home)
  - APPRO P2, 50 (approach 50 mm above part)
  - MOVE P2 (go down to part)
  - CLOSEI (close gripper)
  - DEPART 50 (lift part)
  - MOVE P3 (go to place location)
  - OPENI (release part)
  - END

5. Requirements and Features of Robot Programming Languages

Requirements of a good robot language

- **Easy to learn and use**, readable, English-like commands
- Should be able to define **positions, frames and transformations**
- Should handle **motion control**: speed, acceleration, interpolation and path types
- Should support **sensor data** and decision making
- Should support **real-time** response and interrupts
- Should allow **editing, debugging and testing** without stopping production
- Must provide **safety** features (limits, collision handling, error recovery)

Features of robot programming languages

- **Motion statements**: point-to-point, linear, circular, spline interpolation
- **Data types**: integer, real, boolean, string, joint, location (frame), vector,

array

- **Program flow control**: IF-THEN-ELSE, FOR, WHILE, CASE, GOTO, subroutine call
- **Input / output handling**: digital and analogue I/O, wait-for-signal, interrupts
- **Sensor integration**: vision, force, proximity, tactile
- **Communication**: serial, Ethernet, fieldbus, with PLC and other robots
- **Multi-tasking** (parallel programs) and **error handling**
- **Frames and offsets**: work frame, tool frame, relative moves
- **Simulation and debugging tools**: breakpoints, single step, trace

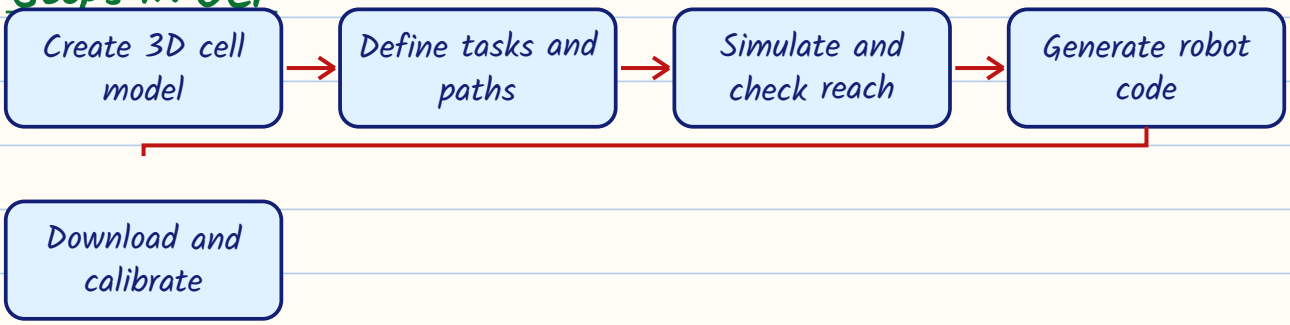
Position = Frame x Offset

A language must let the user write the program once in terms of frames, so that the same program works if the fixture is moved - only the frame is re-taught.

6. Off-Line Programming (OLP) and Software Packages

In **off-line programming** the program is prepared on a computer using a **3D model** of the robot and the cell. The robot is not needed, so production continues.

Steps in OLP



Advantages of off-line programming

- No downtime of the real robot - programming and production in parallel

- Safe: collisions, reach problems and cycle time are checked in the simulation
- Best layout can be chosen before buying equipment
- Programs are stored, copied, edited and reused easily
- Complex paths (welding seams, 3D curves) are generated from CAD data
- Easier integration with CAD / CAM

Disadvantages

- The model never matches the real cell exactly, so **calibration** is required
- Needs trained programmers and costly software
- Errors in the CAD model are carried into the program

Common software packages

Package	Company / use
RoboGuide	FANUC - simulation and OLP
RobotStudio	ABB - simulation and RAPID code generation
KUKA.Sim	KUKA robots
RoboDK	Many robot brands, CAD based
Process Simulate	Siemens - manufacturing cell simulation
DELMIA / IGRIP	Dassault - workcell design
ROS + Gazebo / MoveIt	Open source robot software and simulation

OLP needs calibration, lead-through does not. This is the main trade-off.

7. Problems with Robot Programming Languages

- **No standard** language: every manufacturer has its own, so programs are not portable
- Programmers must learn a new language for every brand
- Limited **sensor and vision** support in many languages
- Difficult to program **complex tasks** such as assembly with force

feedback

- Real-time and multitasking facilities are weak in older languages
- Debugging is poor because the robot moves in the real world, mistakes can cause damage
- Hard to describe 3D geometry in text, hence need for graphic OLP
- **Inaccuracy** of the robot model (link flexibility, backlash, thermal expansion) causes position errors
- Languages do not easily support **error recovery** and exception handling
- Need for trained personnel increases cost

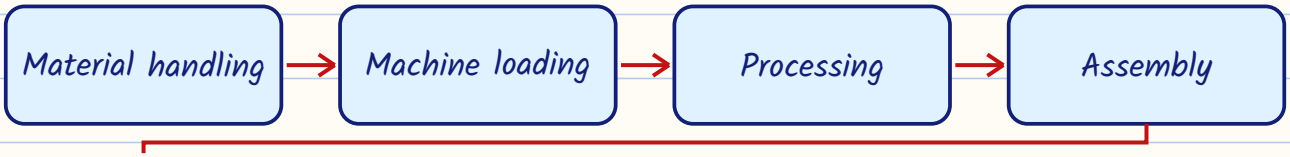
Likely 5 mark question: list the problems of robot languages and suggest solutions.

8. Robot Applications in Manufacturing - Overview

Industrial robots are used where tasks are **repetitive**, **hazardous**, **heavy** or **need high accuracy**.

Main application groups

Group	Examples
Material transfer	Pick and place, palletising, packaging
Machine loading / unloading	CNC lathe, press, die casting, injection moulding
Processing operations	Spot welding, arc welding, spray painting, grinding, drilling
Assembly	Insertion, screwing, fitting, PCB assembly
Inspection	Gauging, vision inspection, testing



Inspection

9. Material Transfer and Material Handling

Material handling: movement, storage, control and protection of materials during manufacture. Robots do the **transfer** of parts from one place to another.

Types of material transfer applications

- **Pick and place**: pick part from one location, place at another (simple 2-4 DOF robots)
- **Palletising and depalletising**: stacking boxes in layers using a defined pattern
- **Packaging**: placing products into cartons
- **Transfer between machines**: along a production line
- **Conveyor tracking**: robot picks parts from a moving conveyor
- **Sorting** with vision

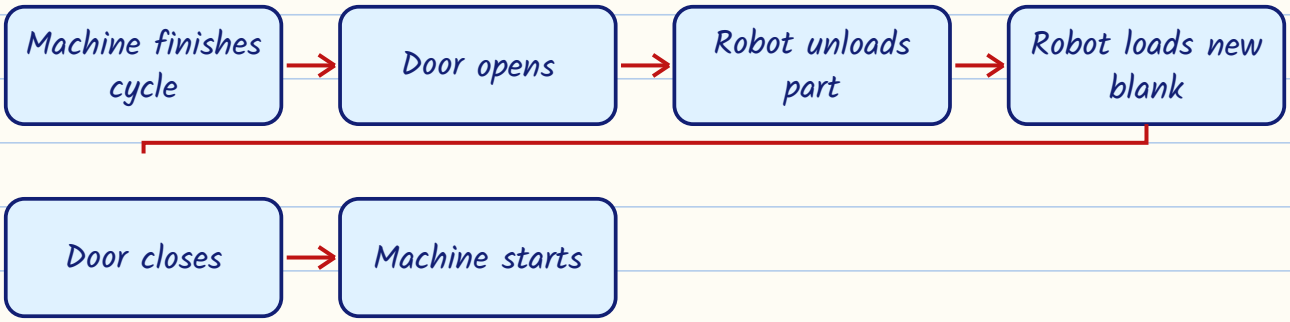
10. Machine Loading and Unloading

A robot **feeds raw parts into a machine** (loading) and **removes finished parts** (unloading).

Applications

- **Machine tools**: CNC lathes and milling machines
- **Press working**: load blanks into presses, remove stamped parts
- **Die casting and forging**: handle hot parts, spray the die
- **Plastic injection moulding**: remove moulded part and runners
- **Heat treatment** furnaces

Robot-machine interface



- Handshake signals: machine READY, door OPEN / CLOSED, part PRESENT, cycle COMPLETE
- Robot waits in a safe position until the signal is received (WAIT instruction)
- A **double gripper** loads a new part and removes the finished one in a single entry

In machine loading the robot is a **slave** of the machine: the machine controls the cycle and the robot responds to its signals.

11. Spot Welding

Resistance spot welding: two sheets are clamped between copper electrodes and heated by a large current for a short time so that they fuse at one spot.

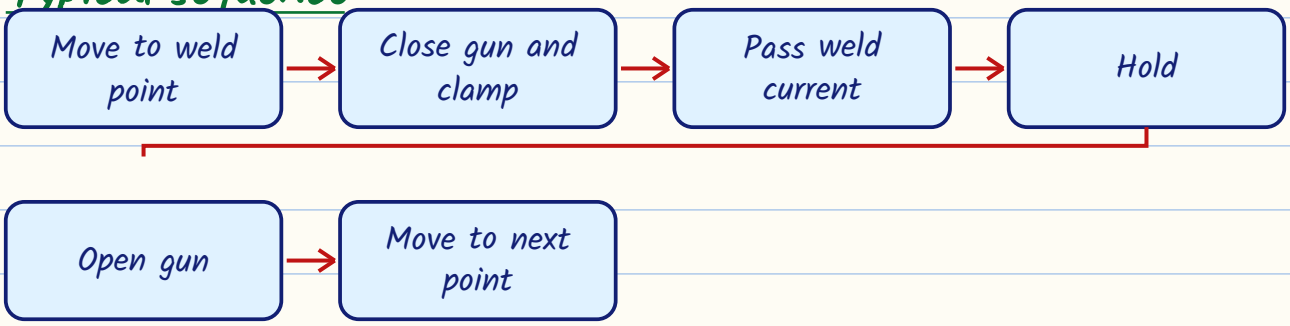
Weld parameters

- Current (5 to 20 kA), weld time (cycles), electrode force and hold time

Robot requirements for spot welding

- **Point-to-point** control with **high repeatability** (about 0.5 mm)
- Payload: weld gun weighs 30 to 100 kg, so a heavy payload articulated robot (6 axes) is used
- Large work volume and **wrist flexibility** to reach the joints in a car body
- Programmed with the **teach pendant**: weld points are taught one by one
- Weld controller interface: start weld, weld complete, gun open / close

Typical sequence



Advantages

- Consistent weld quality, high speed, no operator fatigue or burns

Limitations

- Accessibility of the joint to the gun, heavy weld gun reduces speed

Spot welding = point to point. Arc welding = continuous path. Remember this pair.

12. Continuous Arc Welding

In **arc welding** an electric arc between an electrode (wire) and the workpiece melts the metal, joining the two parts along a **seam**.

Common processes

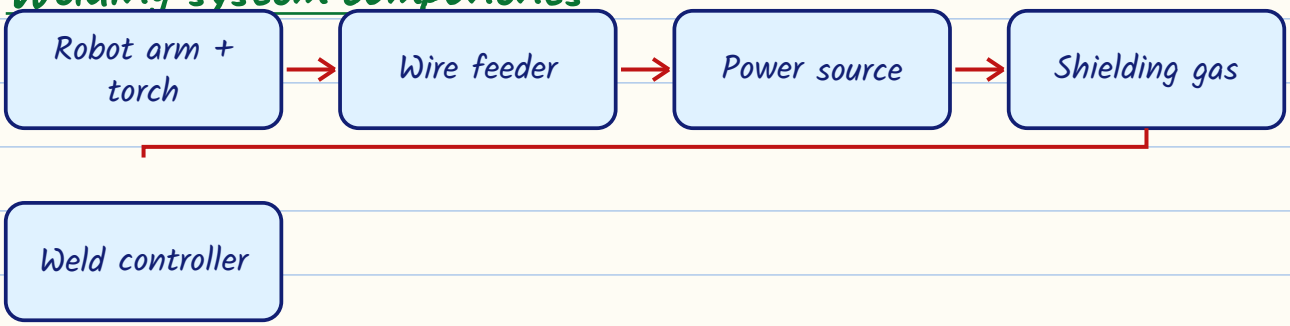
- **MIG / MAG (GMAW)**: continuous wire feed with shielding gas - most common with robots
- **TIG (GTAW)**: tungsten electrode, filler rod, high quality for thin material

Robot requirements

- **Continuous-path** control with straight, circular and weave motions
- Accurate control of **travel speed**, constant torch angle and stand-off distance
- Repeatability around 0.1 mm, payload about 5 to 15 kg
- 6 axes for orientation freedom, often with a **positioner** (2 or 3 axis table) to rotate the work
- Programming: by teach pendant using a few taught points, or manual

lead-through, or OLP from CAD

Welding system components



Advantages

- Uniform weld quality and good penetration, higher arc-on time, safe for the worker (fumes, UV)

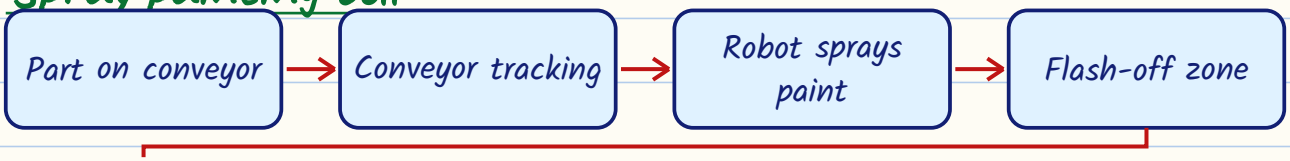
13. Spray Painting

Robots are widely used for **spray painting and coating** because paint fumes are toxic and flammable and quality has to be uniform.

Features and requirements

- **Continuous-path** control with smooth motion and high speed
- Programmed by **manual lead-through** (walk-through): an expert painter moves the arm and the path is recorded and played back
- Explosion-proof design: **hydraulic or intrinsically safe electric** drive, sealed arm, pressurised cabinets
- Large work volume and **many DOF** (6 to 7 axes) to coat complex shapes
- Memory capacity high due to the large number of stored path points
- Control of paint flow, atomising air, gun trigger and colour change

Spray painting cell



Oven curing

Advantages

- Uniform coat thickness, less paint waste (saves 20 to 30 percent paint)
- Operator protected from fumes and noise
- Fast colour change, consistent finish
- Reduced ventilation costs

Painting robots avoid electric sparks: explosion-proof enclosure with positive air pressure is mandatory.

14. Assembly

Assembly is about 40 to 50 percent of the manufacturing cost, so automation here is valuable.

Robots for assembly

- **SCARA robots**: high speed, compliant in the horizontal plane, ideal for vertical insertion (PCB, small parts)
- **Cartesian robots**: accurate and rigid, for simple assembly
- **Articulated robots** and dual-arm robots: complex assembly

Assembly tasks

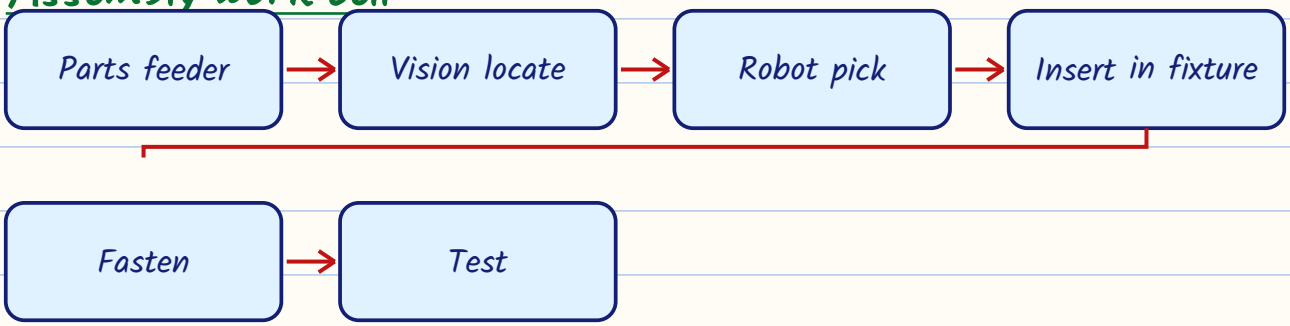
- Part insertion (peg in hole), screw fastening, snap fit
- Adhesive application, riveting, press fit
- PCB component insertion (electronic assembly)

Requirements

- **High repeatability** (0.01 to 0.05 mm) and moderate speed
- Compliance for insertion: **RCC (Remote Centre Compliance)** device to correct small errors
- Force / torque sensing for fitting parts
- Vision to find part orientation, fixtures and feeders to supply parts

- Compact, light grippers, quick tool changers

Assembly work cell



> SCARA = Selective Compliance Assembly Robot Arm - the standard assembly robot.

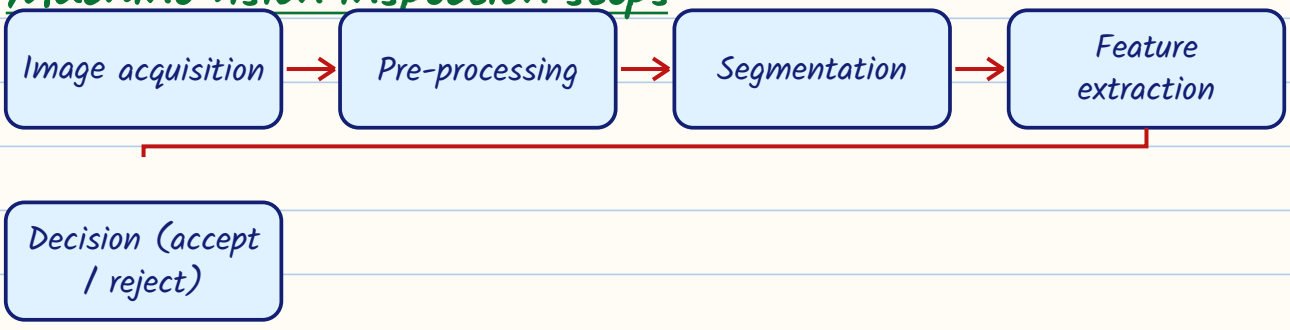
15. Inspection

Robot inspection checks parts for **dimensions**, **defects**, **presence of components** and **surface quality** automatically.

Inspection methods

- **Contact inspection**: touch probe or gauge mounted on the robot - used like a CMM
- **Non-contact inspection**: vision camera, laser scanner, ultrasonic, eddy current, X-ray

Machine vision inspection steps



Advantages

- 100 percent inspection at line speed, objective and repeatable results
- Data recorded for **statistical process control** (SPC)

- Safe in hazardous places such as nuclear plants

Limitations

- Lighting and part presentation affect results
- Robot accuracy is less than a dedicated CMM
- Calibration needed

16. Summary and Quick Revision

Programming

Topic	Key points
Lead-through	Powered (pendant, PTP), manual (walk-through, CP)
Textual languages	VAL, AML, AL, RAPID, KRL; branching, sensors, subroutines
Off-line	3D simulation, no downtime, needs calibration
Problems	No standard, poor sensor support, debugging risk

Important exam questions

- Explain the methods of robot programming with neat sketches
- Compare powered lead-through and manual lead-through
- List the requirements and features of a robot programming language
- Write a VAL program for a pick and place task
- Describe off-line programming and its advantages and disadvantages
- What are the problems with robot programming languages?
- Discuss the robot requirements for spot welding and arc welding
- Explain the use of robots in spray painting
- Explain robot machine loading and unloading with an example
- Write short notes on robot assembly and inspection

Last minute tip: for every application write (1) task, (2) type of control PTP or CP, (3) robot type and features, (4) advantages. This scores full marks.